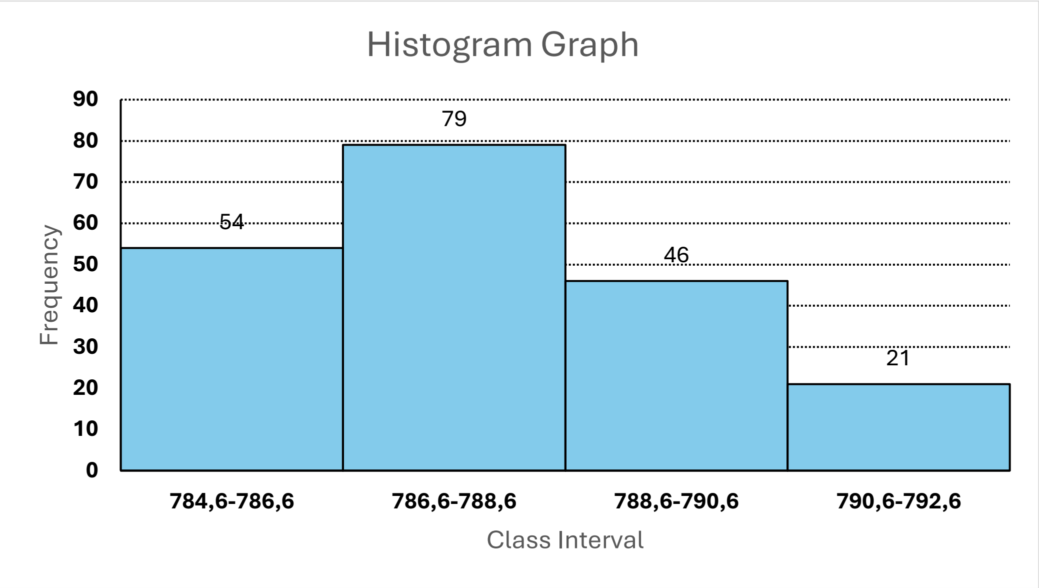
FUNDA GÜÇLÜ 23058653

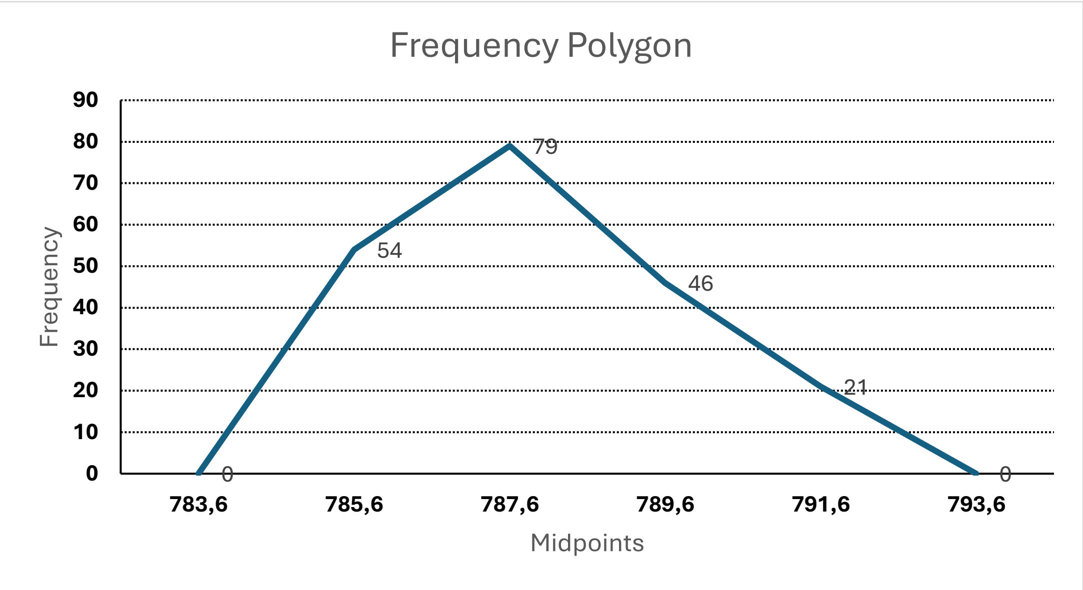
Question 1:

class interval	midpoints	frequencies	increasing cumulative frequencies	decreasing cumulative frequencies
784,6-786,6	785,6	54	54	200
786,6-788,6	787,6	79	133	146
788,6-790,6	789,6	46	179	67
790,6-792,6	791,6	21	200	21

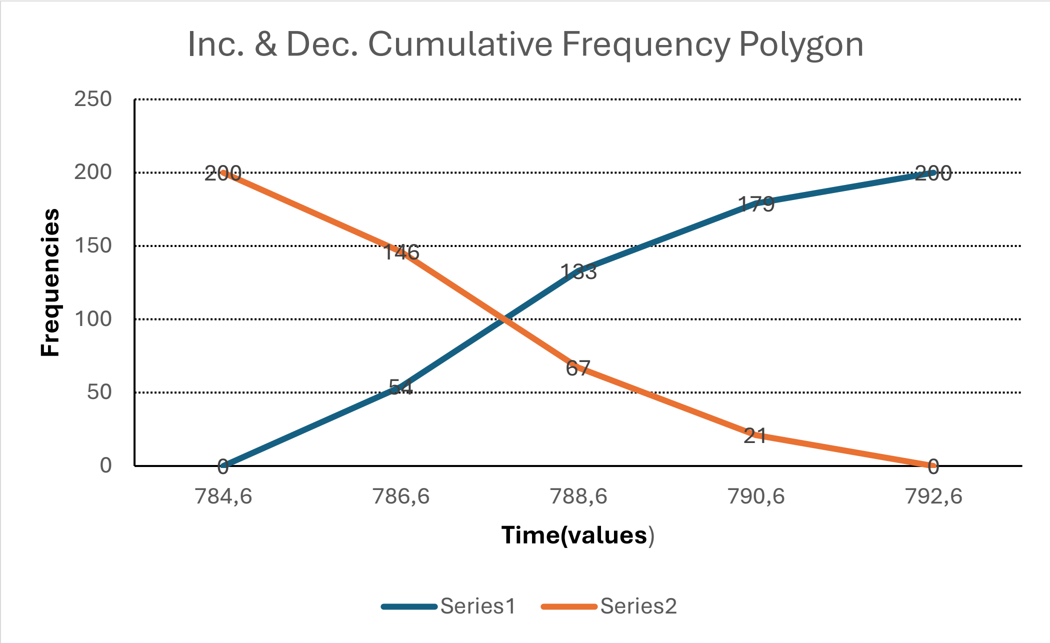
inc. cumulative relative freq.	dec. cumulative relative freq.	relative freq.
0,27	1	0,27
0,665	0,73	0,395
0,895	0,335	0,23
1	0,105	0,105

Question 2:





Question 3: Series1=Inc. Polygon Series2=Dec. Polygon



Question 4:

Mean (arith.):	Median:	Mode:	Standart Deviation:
788,008	787,7	786,4	1,774777733

Mean<Median<Mode => skewed to left

Mean>Median>Mode =>skewed to right

Mean=Median=Mode =>symmetric

According to our values our graph is skewed to right.

Question 5:

Mean (arith.):	Median:	Mode:	Standart Deviation:
787,94	787,764557	787,462069	1,887961864

Question 6:

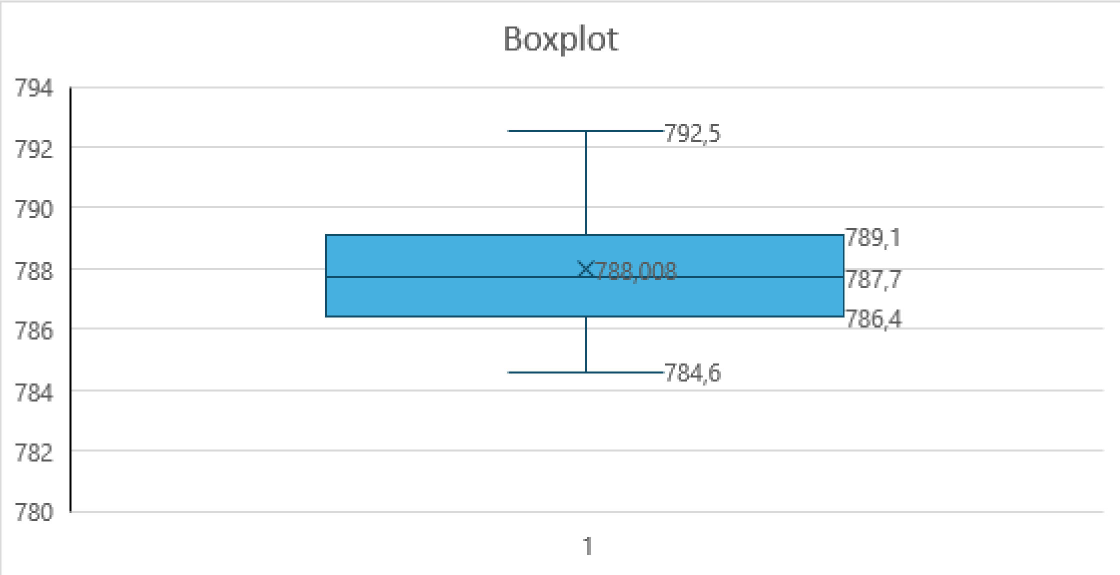
Quartile1:	Quartile2:	Quartile3:	Interquartile Range:
786,4	787,7	789,1	2,7

|Median-Q3|>>|Median-Q1| => Data is skewed to right

|Median-Q3|<<|Median-Q1| => Data is skewed to left

|Median-Q3|≈|Median-Q1| => Data is symmetric

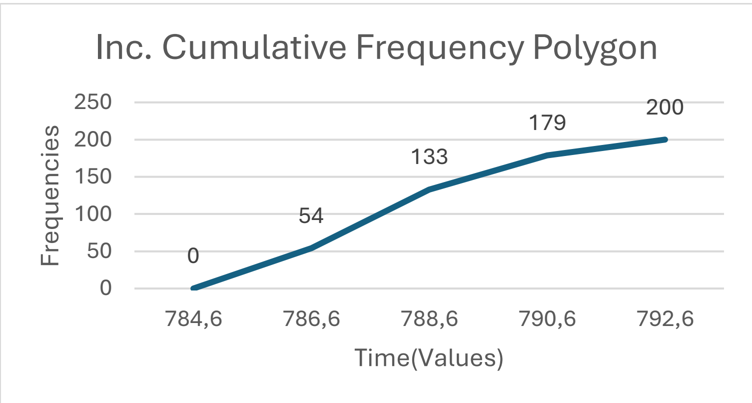
According to our values our graph is skewed to right.



Question 7:

# freq. İn μ-(σ/2):	
	69,64275319
# freq. İn μ+(σ/2):	
	139,5315614
# data in inc. cu. freq. poly:	
	69,88880825
Zlower:	
	0,308537539
Zupper:	
	0,691462461
P(Zlower<Z<Zupper):	
	0,382924923
# data in normal distribution:	
	76,58498451
diff. Norm. Dist& Freq. Poly.:	
	6,696176259
6,696176259≈ 7	

Mean:	**μ+(σ/2)**
787,94	788,8839809
Std. Deviation:	**μ-(σ/2)**
1,887961864	786,9960191



Question 8:

A)

Mean:	Standart Dev.:	Variance:	Sample size:
788,008	1,779231385	3,165664322	200

B)

The population variance is unknown. We know the sample's size is 200(n>=30)

mean formula is:

$$CI = \bar{x} \pm z\frac{s}{\sqrt{n}}$$

α = 0,08	**lower bound of CI:**
α/2 = 0,04	787,787745
$Z\alpha/2$:	**upper bound of CI:**
1,750686071	788,228255
mean=788,008	**Confidence interval is:**
n=200	[787,787745 - 788,228255]
square root of n:	
14,14213562	

C)

$$\left[\frac{(n-1)S^2}{\chi^2_{\frac{\alpha}{2},n-1}}, \frac{(n-1)S^2}{\chi^2_{1-\frac{\alpha}{2},n-1}}\right]$$

$\chi^2_{\alpha/2}$ value is 235,2635115

$\chi^2_{1-\alpha/2}$ value is 165,4880149

α = 0,08
α/2 = 0,04
n=200
n-1=199
variance:
3,165664322

lower bound of CI:
2,677708906
upper bound of CI:
3,806724012
Confidence interval is:
2,677708906 - 3,806724012

D)

X:
776
H0:μ=11,6+X
787,6
Hα:μ≠ 11,6+X
α = 0,08

The population variance is unkown.

We know sample size=200>=30

$$z_0 = \frac{\bar{x} - \mu_0}{S/\sqrt{n}}$$

square root of n:	P value=2*P(z>z0):
14,14213562	0,001182913
Standart Deviation:	
1,779231385	
sample mean:	P value = 0,001183< α = 0,08
788,008	We reject H0. (Pvalue<α)
Z0:	
3,242968499	

E)

H0:μ>=12,2+X
788,2
Hα:μ< 12,2+X

The population variance is unkown. We know sample size=200>=30

$$z_0 = \frac{\bar{x} - \mu_0}{S/\sqrt{n}}$$

square root of n:	Pvalue=P(z<z0):
14,14213562	0,06349213
Standart Deviation:	
1,779231385	
sample mean:	
788,008	
Z0:	
-1,526102823	

To reject H0 α must be larger than P value.
Pvalue= 0,06349213 so for all values for α>0,06349213
we reject H0.
(α should be < 1)